HIC02060: An Imperfect but Targeted Drug for Ventricular Tachycardia caused by the high EZ level

Introduction

High EZ level caused by rare, small benign peritoneal tumors can lead to ventricular tachycardia (VT), possibly resulting in cardiac death Since the tumors cannot be removed by chemical or surgical treatment, a pharmacological therapy is needed. As a neurotransmitter or a neurohormone, EZ has two receptors, namely EZ1R and EZ2R, whereas only EZ's binding to EZ1R causes VT, the drug HIC02066 (HIC) targeted the binding is introduced to treat the disease and be assessed in this report according to the experiment data. This report will analyze the binding properties of the drug, investigate into the potential effects of the drug, discuss whether the drug reach the expectation of avoiding being overly released into the circulation on normal cardiac function. and lastly provides insights into further research directions.

Properties of EZergic Receptors and HIC02066

To understand the mechanism of the drug's effect, the detailed research into the chemical properties of the drug much be carried out. Based on Dataset 1, we can graph and visualize the binding of EZ and HIC at EZ1R and EZ2R respectively (Rang, et al., 2012, pp. 6-21). With the help of R studio (Appendix 4), each binding curve was trained to fit a sigmoid curve by using non-linear least squares method, outputting an approximation of the concentration required for 50% effect (EC_{50}) and the maximum effect (E_{max}) (Appendix 1.2) (Bates & Chambers, 1992; Bates & Watts, 1988).

Calculating and comparing the affinity

The affinity of the drug and EZ to the two receptors is important for us to understand the pharmacodynamic. The affinity of a ligand is expressed as the association constant (K_a), which inversely proportional to the disassociation constant (K_d). According to Hill Equation, $\text{Occupation Percentage} = [\text{Ligand}]/(K\_d + [\text{Ligand}])$. (Gesztelyi, et al., 2012) Therefore, if the occupation percentage = 50%, $K_d = [\text{Ligand}]$; therefore, we can calculate the K_d in this way. Hence, we can read the K_d from Dataset 1. The affinity of the drug at EZ1R and EZ2R is 3.04E+06, 1.63E+06, respectively, while the affinity of EZ at EZ1R and EZ2R is 1.10E+06, 1.63E+06, respectively (Appendix 1.2).

HIC can theoretically bind to EZ1R more specifically due to two reasons. First, the affinity that EZ and HIC have the same affinity in binding EZ1R, while HIC02066 have a much stronger affinity to EZ1R than EZ, which can help the drug to specifically bind to the targeted EZ1R. Second, the affinity of either EZ or HIC02066 to EZ2R is higher than EZ's affinity to EZ1R and lower than HIC02066's affinity to EZ1R; thus, EZ tends to bind to EZ2R, while HIC02066 tends to bind to EZ1R.

The activity of HIC02066 in binding to ezernergic receptors

EZ is the natural agonist of EZ1R, EZ2R, whereas HIC was designed to downregulate EZ's binding to EZ1R with its high affinity to EZ1R. To visualize the effects of HIC02066 on cAMP level, we fitted a curve to the Dose-Response relation of the drug when binding EZ1R given different drug concentrations (Appendix 2). In HIC's binding to EZ1R, EC50 remains unchanged, yet the maximum effect decreased after adding the drug, which are the hallmark of irreversible antagonist helping us confirm the role of EZ at EZ1R.

EZ and HIC may well both be the reverse agonists of EZ2R, instead of a modulator. The Schild regression of the dose-response curve of HIC at EZ2R shows a linear relation, which indicates the competitive relationship between HIC and EZ2R but not Kd of the drug at EZ2R (Kenakin, 1982; Wyllie & Chen, 2007). According to the dose-response graph (Appendix 3), either the increased HIC level or the increased EZ level can lead to the inhibition of the response. Furthermore, EZ and HIC have the same affinity to bind to EZ2R; despite the competitive relation, they will bind to the same amount of EZ2R.

Pathology and the target receptors

Cause of the disease and the harm of high EZ

To understand why our researchers think targeting EZ receptors may help, we need to understand the pathology of the ventricular tachycardia and figure out what is the possible target of the disease. In the patients, EZ is overly produced by the peritoneal tumors, which negatively affects the cardiac health. The devastating results of the exceeding EZ has been confirmed in our animal models. In our 8-week follow-up experiment, all the male mice and most of female mice administered with only exogenous EZ to mimic the tumor's overexcretion of EZ (exEZ group), had been dead before the experiment ended. In contrast, most mice in 3 other groups of 10 male and 10 female mice survived, including those administered with HIC alone (HIC group), those administered with both HIC and exogenous EZ (exEZ +HIC group), and the control group (PR Dec 2018: Figure 1).

Classification of the receptors

All the receptors we targeted in the experiment can be EZenergetic receptors according to the ligand. EZ as a neurohormone could regulate part of the pathway of the cardiac muscle contraction; or else, it would be better to come to curative therapies including catheter ablation (Dukkipati, et al., 2017). Since ionotropic and metabotropic receptors are two major receptors of neurotransmitters, EZ1R and EZ2R could be ionotropic or metabotropic receptors (Ferré, et al., 2007). Previous studies show that EZ1R is mainly expressed throughout the cardiac muscle while EZ2R is mainly expressed on endothelial cells of the afferent arterioles of the glomeruli of the kidney, where the Renin-Ang system is involved and regulates the cardiovascular system (Peti-Peterdi & Harris, 2010).

For ion channels play an important role in activation and deactivation of cardiac functions (Nattel, et al., 2007) and has a role in renal function, it is not unlikely that EZernergic receptors are ionotropic receptors that directly act on ion channels and cause a sequent change in the energy and calcium equilibrium throughout the body. Yet, the constitutive activity-like curve of the drug dose-response curve (Appendix 2&3) reminds us of the facts the natural high constitutive activity of GPCRs and GPCRs induce the production of cAMP (Milligan, 2003). Furthermore, cAMP can influence the ion channels via the calcium pathways and so on (Rang, et al., 2012). Considering the complex experiment's results involving both reduced renal function and protected heart function, it is more likely that EZ1R and EZ2R regulate the heart and kidney via a signaling networks where GPCRs can exhibit flexibility of signaling (Vaidehi, 20.

Considering the possibility of the multi-role compound and the conflicting effects of HIC binding (Appendix 2&3), a research into the pharmacokinetics should be performed to confirm the role of these receptors.

Adverse effects of the HIC therapy

Admittedly, HIC has been confirmed to have a protective effect under high EZ, for most mice with high EZ and treated with HIC survived the experiment while those not treated with EZ died. However, the application of HIC can cause reduced renal function. People with impaired renal function should not use the drug, especially for adult males, for male mice treated with HIC alone have still shown the highest GFP and TUP, which indicates impaired renal function then.

Male mice seemed to suffer from more inverse effects. The male EZ group suffered from deaths since the first week and the whole group did not survive the sixth week, while female group began seeing death in the third week. (PR Dec 2018: Figure 1),although in the 4th weeks of the experiment, both female and male exEZ group began seeing tachycardia patients (PR Dec 2018: Table 1).

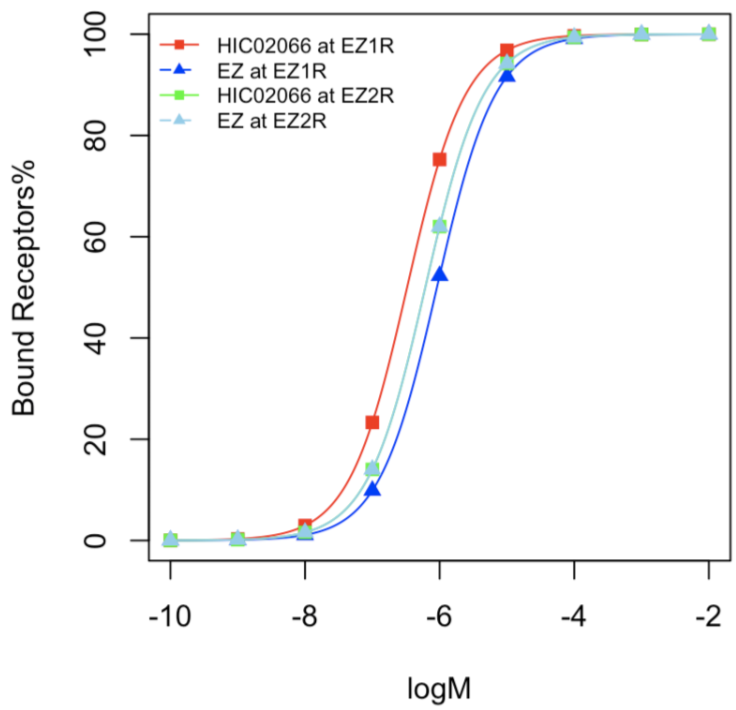
High EZ mice administered with HIC seemed suffering from more renal function loss, since mice in the group had the highest number of impaired glomeruli and signs of glomerular injury, repair and fibrosis(PR Dec 2018: Figure 3&4) (Krenning, et al., 2010). Thus, the renal function should be constantly checked for any to take a long time administration of the drug.

Conclusion: Outlooks for the Treatment

HIC has exhibited a protective effect on sustaining the cardiac function and avoiding tachycardia, although the effects do not come without a price. The administration of HIC via mini-pumps can lead to a reduced renal function. The high EZ level added HIC therapy can further worsen the reduction. Even so, the adverse effects are still far less lethal than the cardiac events caused by the high EZ level. For those troubled by high EZ, this drug is one of rare drugs that is targeting the causes of the pathogenesis, which could reduce the side effect when taking other medicine.

For future research, first, a toxicity test should be done to define the safe dosage. Second, the worsened situation for the mice administered with both HIC and EZ needs further research and a solution. Third, if preparing for human testees after the previous two questions have been solved, improvement of administration will be needed. Lastly, biological availability and other important data for drug development is so lost for this drug and needs further investigations.

Appendix 1 Binding graph of EZ and ezernergic receptors & HIC020266



Appendix 1.1 Parameters of curve fitting of the binding data from Dataset 1

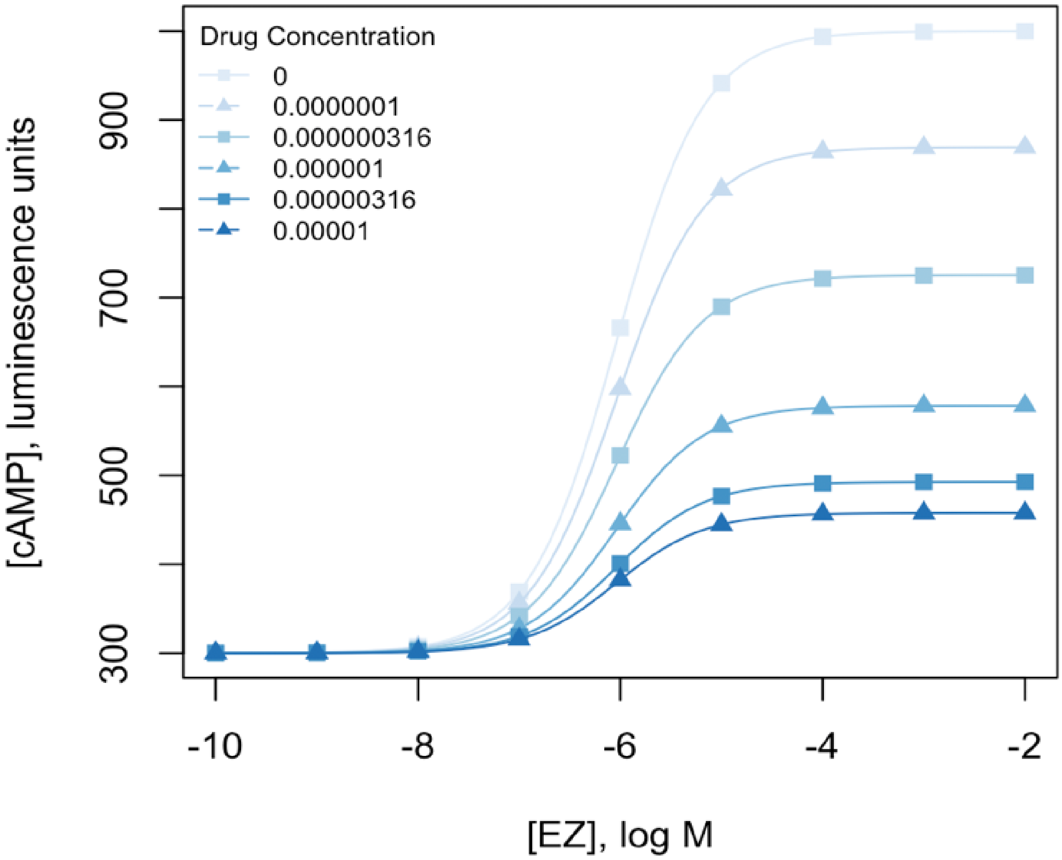
	EZ2R-EZ	EZ2R-HIC	EZ1R-EZ	EZ1R-HIC
Max Occupation (%)	1.00E+02	1.00E+02	1.00E+02	1.00E+02
Conc. for 50% occupation	6.13E-07	6.13E-07	9.12E-07	3.29E-07
Residual sum-of-squares (RSS)	7.47E-05	7.47E-05	3.78E-05	4.39E-05

Appendix 1.2 Affinity calculation

	EZ2R-EZ	EZ2R-HIC	EZ1R-EZ	EZ1R-HIC
EC50	6.13E-07	6.13E-07	9.12E-07	3.29E-07
Affinity	1.63E+06	1.63E+06	1.10E+06	3.04E+06

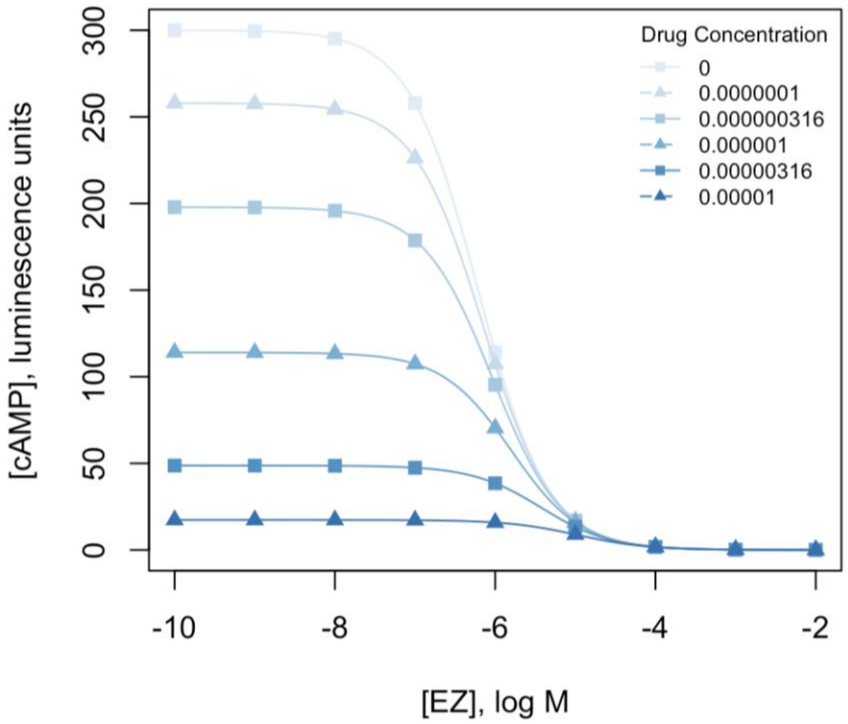
	HIC	EZ
EZ2R	1.63e+06	1.63e+06
EZ1R	3.04e+06	1.1e+06

Appendix 2 Curve fitting in Dataset 2



	[Drug]= 0	[Drug]= 1e-7	[Drug]= 3.16e-7	[Drug]= 1e-6	[Drug]= 3.16e-6	[Drug]= 1e-5
EMax	7.00e+02	5.694e+02	4.254e+02	2.785e+02	1.927e+02	1.578e+02
EC50	9.12e-07	9.121e-07	9.121e-07	9.118e-07	9.125e-07	9.130e-07
RSS	0.005914	0.009925	0.00773	0.004457	0.004737	0.00246

Appendix 3 Curve fitting in Dataset 3

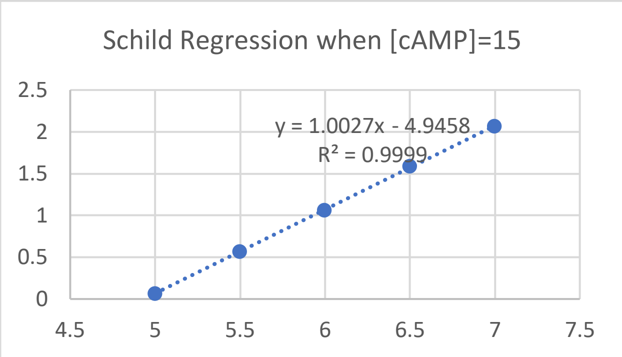


	[Drug]= 0	**[Drug]= 1e-7**	**[Drug]= 3.16e-7**	**[Drug]= 1e-6**	**[Drug]= 3.16e-6**	**[Drug]= 1e-5**
E_{Max}	3.000e+02	2.579e+02	1.979e+02	1.140e+02	4.871e+01	1.732e+01
EC_{50}	1.632e-06	1.403e-06	1.076e-06	6.201e-07	2.648e-07	9.425e-08
RSS	5.32e-05	7.418e-05	7.055e-05	4.906e-05	5.694e-05	6.769e-05

Appendix 3.1 Schild Regression

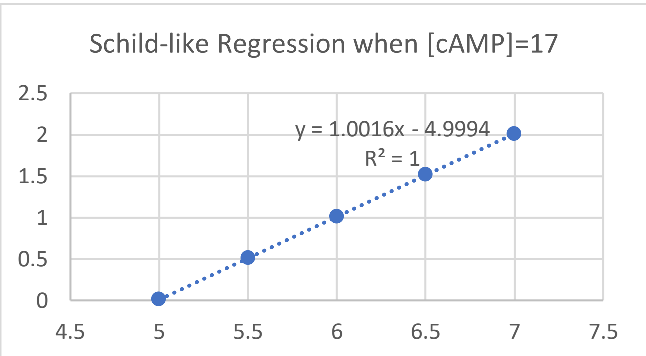
When [cAMP]=15,

[Drug]	A	-log[Drug]	-log(1-A'/A)
0	1.16E-05		
1.00E-07	1.15E-05	7	2.064457989
3.16E-07	1.13E-05	6.5	1.587336735
1.00E-06	1.06E-05	6	1.064457989
3.16E-06	8.48E-06	5.5	0.570303395
0.00001	1.64E-06	5	0.066198651



When [cAMP]=17

[Drug]	A	-log[Drug]	-log(1-A'/A)
0	1.02E-05		
1.00E-07	1.01E-05	7	2.008600172
3.16E-07	9.89E-06	6.5	1.517238478
1.00E-06	9.20E-06	6	1.008600172
3.16E-06	7.04E-06	5.5	0.508913089
0.00001	2.03E-07	5	0.00873048



When [cAMP]=…, et cetera

Appendix 4 Source code of the curve fitting

Environment: RStudio Version 1.1.463 /   R 3.5.2 GUI 1.70 El Capitan build (7612)
Exemplar Source Code:

```
rawdata<-read.csv("Compete.csv",header=TRUE)
#data has been processed.
n=9
color='#DEEBF7'
df<-data.frame(rawdata[,1],rawdata[,n])
my_data = df
Dose<-df[,1]
Response <- df[,2]
my_max <- 300
my_EC50<- 1e-6
fitmodel <- nls(Response~Max * 10^(-12-Dose)/(EC50 + 10^(-12-Dose)), data=my_data,
start=list(Max = my_max, EC50=my_EC50))
#this fitting needs adjustment from time to time, check here before use the model.
fitmodel
params <- coef(fitmodel)
Max <- params[1]
EC50 <- params[2]
X2 <- seq(-2,-10, -0.01)
Y2 <- Max * 10^(-12-X2)/(EC50 + 10^(-12-X2))
#visualisation
plot(Y2~X2,type="l", ylim=c(0,300), xlab="[EZ], log M", ylab="[cAMP], luminescence
units", col=color)
points(Response~Dose, data=my_data,col=color,pch=15)
```

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